

ARAVIND KRISHNAKUMAR

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PERSONAL DATA

Place | Date of Birth: Kerala, India | 11 March 1992
Address: 49/728K Krishnageetham, Elamakkara,
Ernakulam, Kerala, India, Pin: 682026
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EDUCATION

July 2017 - July 2022 **PhD in Astronomy & Astrophysics**, PRL, Ahmedabad, India
Thesis title - "OBSERVATIONAL ANALYSIS OF COMETARY BODIES IN THE SOLAR SYSTEM."
July 2013 - May 2015 **Master of Science in Physics**, Loyola College, Chennai, University of Madras, India
Thesis: "WET CHEMICAL SYNTHESIS AND CHARACTERIZATION OF CDS NANOPARTICLES"
Advisor: Assistant Professor Dr. N. S. Nirmala JOTHI
GPA: 8.29/10
July 2010 - June 2013 **Bachelor's in Science with honours in Physics**, Sacred Heart College, Thevara,
Mahatma Gandhi University, Kottayam, Kerala
GPA: 3.49/4
June 2010 **Class XII** Central Board of Secondary Education, Kerala, India
PERCENTAGE: 89/100
June 2008 **Class X** Central Board of Secondary Education, Kerala, India
PERCENTAGE: 92/100

AWARDS AND FELLOWSHIPS

Nov. 2024 - Present WBI Fellow, Post-Doctoral Researcher, University of Liege, Belgium
Sept. 2022 - Sept. 2024 Post-Doctoral Fellow, Physical Research Laboratory, Ahmedabad
July 2023 Ph.D in Physics, Indian Institute of Technology, Gandhinagar, India
July 2019 - July 2022 Senior Research Fellow, Physical Research Laboratory, Ahmedabad
2017 - July 2019 Junior Research Fellow, Physical Research Laboratory, Ahmedabad
2017 Qualified for Junior Research Fellowship (JRF) in CSIR-UGC NET JUNE 2017
2017 Qualified Joint Entrance Screening Test (JEST), All India Rank : 236
2017 Qualified GATE exam in Physics, All India Rank : 419
2016 Qualified for Lectureship in CSIR-UGC NET DEC 2016
2013 - 2015 GOLD MEDAL - M.Sc. Physics Batch topper, Loyola College, Chennai
2013 - 2014 GOLD MEDAL - M.Sc. Physics 1st and 2nd Semester batch topper, Loyola College, Chennai

TECHNICAL SKILLS

Programming Languages and Softwares : Basic Knowledge in C++, C, HTML and proficient in IRAF, PYTHON, $\text{\LaTeX}$, Beamer and esoreflex
Telescope Observations : Hands-on experience with 3.6 m, 2.5 m, 2 m, 1.2 m, 50 cm and 43 cm telescopes.
Accepted Telescope proposals: Multiple proposals on the 1.2 m and 2 m Telescopes. Two proposals for the 3.6 m telescope and two with the space telescope ASTROSAT (first comet observation on both facilities). One Rank B accepted proposal in the Subaru telescope for mid-IR observation of comets with the COMIC instrument during service period. Two proposals on the Observatoire de Haute Provence (OHP) with 4 nights allocated.

Instruments Used : LISA Spectrograph, Optical CCD imagers, NICSPol, Optical Polarimeter, HFOSC(Low-res), HESP(High-res), ADFOSC(Low-res), TANSPEC(NIR), UVIT.

Astronomical observations performed : Spectroscopy, Photometry and Polarimetry of comets.

Computer OS : Ubuntu, macOS, Windows

RESEARCH EXPERIENCES

- **MSc course:** Project titled “WET CHEMICAL SYNTHESIS AND CHARACTERIZATION OF CDS NANOPARTICLES”, under the guidance of Dr N. S. Nirmala Jothi, Assistant Professor, Loyola College.
- **PhD course work 1st semester:** Project titled “STABILITY ANALYSIS OF OPTICAL SPECTROGRAPH USED IN THE 1.2M TELESCOPE AT MT.ABU”, under the guidance of Dr Shashikiran Ganesh, Physical Research Laboratory, Ahmedabad. This project involved the extensive use of Image Reduction and Analysis Facility (IRAF) to reduce and analyse spectroscopic data to get a rough picture of the stability of the telescope and detector during observations.
- **PhD course work 2nd semester:** Project titled “EXTRACTION AND FLUX CALIBRATION OF HANLE FAINT OBJECT SPECTROGRAPH CAMERA (HFOSC) DATA”, using IRAF, under the guidance of Dr Shashikiran Ganesh, Physical Research Laboratory, Ahmedabad.
- **PhD summer semester:** Project titled “MEASURING STAR BRIGHTNESS : APERTURE PHOTOMETRY”, using self-developed Python code under the guidance of Dr Vishal Joshi, Physical Research Laboratory, Ahmedabad.
- **PhD summer semester:** Project titled “EVALUATING THE THICKNESS OF ASTROCHEMICAL ICE ANALOGS” under the guidance of Dr Bhalamurugan Sivaraman, Physical Research Laboratory, Ahmedabad.
- **PhD Thesis:** OBSERVATIONAL ANALYSIS OF COMETARY BODIES IN THE SOLAR SYSTEM. Comets, the most primordial remnants of our Solar system, can provide us with immense details regarding the conditions that prevailed during the initial stages. Indian telescopes were used to perform low-resolution spectroscopy of 22 comets (including comets from different reservoirs and the interstellar comet 2I/Borisov). High-resolution spectroscopic, imaging, and polarimetric observations were also conducted on a few comets to study these bodies containing pristine volatile materials. The results extracted from these observations were used to learn more about the molecules present, their production rate, column density, dust-to-gas ratio, dust grain parameters, Ortho-to-Para ratio of NH_2 , Green-to-Red doublet ratio of forbidden oxygen line, etc. Further information like the relative molecular abundance or line ratios and their behaviour with heliocentric distances was extracted to realise the individual comet’s composition and to get a collective understanding of the variations.
- **Post-Doctorate:** During the initial postdoctorate, further comets were followed in optical imaging and spectroscopy (low- and high-resolution). To extend expertise into different wavelength regimes, the TANSPEC instrument mounted on the 3.6 m DOT telescope in India was used to study comets in the IR regime. Although the inaccessibility of windows above $2.5\ \mu\text{m}$ restricts us from directly observing the emissions from molecular gases, these observations were used to analyse the continuum emission to probe the dust particles in detail. The possibility of using the existing observed data set to detect hidden ionic emissions was explored and succeeded for the comet C/2020 F3 (NEOWISE). The possibilities of computing a new set of scale lengths of the parent and daughter for different molecules were also explored and are being continued.

During my current postdoctoral research, I intend to implement a new approach to study in great detail, and within the same team, the chemical composition of a few comets in the coming year using IR, near-UV and visible simultaneous observations with state-of-the-art new IR instrumentation available now to European astronomers (CRIRES+ at the VLT). This approach will reduce inconsistencies and systematic differences associated with the diverse reduction and analysis techniques. Furthermore, the simultaneity that we propose between the IR and optical will take into account the fact that comets are variable objects but rarely treated as such when considering their composition. With this approach, we will develop new expertise to better understand comets’ coma composition and physical processes.

CONFERENCES/SEMINARS/SUMMER SCHOOLS ATTENDED

- May 2014:** Introductory Summer School on Astronomy and Astrophysics conducted by Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.
- December 2014:** Workshop on ‘RECENT DEVELOPMENTS IN COSMOLOGY’ conducted by IUCAA Resource Centre (IRC). Kochi, India.
- September 2018:** Attended National level students conference, Young Astronomers Meet (YAM), Physical Research Laboratory (PRL), Ahmedabad, India.
- August 2019:** Selected for 5TH INDO-FRENCH ASTRONOMY SCHOOL on *Spectroscopy and Spectrographs* to be held in IUCAA, Pune, India from AUGUST 16 - 24, 2019.
- February 2020:** Delivered a talk titled ‘CONTRASTING BEHAVIOUR OF TWO JUPITER FAMILY COMETS’ at the 38th Meeting of the Astronomical Society of India held at IISER Tirupati, between 13th - 17th February 2020.
- August 2020:** Delivered a talk titled ‘*Spectroscopic and Imaging study of the first interstellar comet 2I/Borisov*’ at the International Conference on Dust in Astrophysics held at Assam University, between 31st August - 1st September 2020.

- September 2020:** Presented a poster titled '*Spectroscopic and Imaging study of first Interstellar comet 2I/Borisov from two Indian Observatories*' at the Poster exhibition held by the Royal Astronomical Society (RAS) on, 14th September 2020.
- February 2021:** Presented a poster titled 'SPECTROSCOPY, POLARISATION AND DUST MODELLING OF *short period comet 156P/Russel-Linear*' at the Indian Planetary Science Conference (IPSC-2021), which was organised online during 25-26 February 2021.
- June 2021:** Invited to deliver a talk on the '*Minor bodies in the Solar system*' by the Department of Physics, School of Applied Sciences, REVA University, Bangalore, during the occasion of 'World Asteroid Day 2021' on June 30th, 2021.
- March 2022:** Presented a poster titled '*Optical polarimetric study of cometary dust*' at the 40th Meeting of the Astronomical Society of India held at ARIES, Nainital, between 25th - 29th March 2022.
- September 2022:** Delivered a talk titled '*Polarimetric observation of cometary bodies*' at the 7th SCOP conference held in PRL, Ahmedabad between 28th - 30th September 2022.
- September 2022:** Delivered a talk titled '*High resolution optical spectroscopic comparison of a short period and long period comet*' at the Europlanet Science Congress 2022, held in Granada, Spain. The abstract was withdrawn later due to travel budget issues.
- March 2023:** Delivered a talk titled '*Peculiar long period comet C/2019 L3 observed from Indian and Belgian facilities*' at the 41st Meeting of the Astronomical Society of India held at IIT Indore, between 1st - 5th March 2023.
- March 2023:** Delivered a talk titled '*Optical spectroscopy of comets*' at the 3rd BINA conference held at Graphic era Hill University, hosted by ARIES, Nainital, between 22nd - 24th March 2023.
- June 2023:** Presented a poster titled '*Revealing the ionic emissions in comet C/2020 F3 (NEOWISE)*' at the 14th ACM conference held at Flagstaff, Arizona, between 18th - 23rd June 2023.
- September 2023:** Presented a poster titled '*Polarisation: Probing the dust particles of Solar system bodies*' at the 3rd Venus-SC conference held at PRL, Ahmedabad, between 21st - 22nd September 2023.
- October 2023:** Invited to deliver a talk titled '*Unlocking the Mysteries of Comets: The Significance of Long-Term Monitoring through Spectroscopy and Photometry in Indian-Belgian Collaboration*' at the 1-day BINA/BIPASS meeting held in Brussels on 10th October 2023.
- August 2024:** Delivered a talk titled '*Unveiling Cometary Composition: Importance of Spectroscopic Follow-up in the Era of Large Discovery Surveys*' at the 32nd IAU General Assembly held at Cape Town, South Africa, between 6th - 15th
- September 2025:** Delivered a talk titled '*Investigating the Molecular Activity and Composition of Comet C/2022 E3 via Photometry and Optical Spectroscopy*' at the Europlanet Science Congress 2025, held in Helsinki, Finland.
- November 2025:** Delivered a talk titled '*Tracing Comet Evolution: Insights from Multi-Site Observations of Comet C/2020 F3 (NEOWISE) and C/2022 E3 (ZTF)*' at the 4th BINA conference held at University of Calicut, Kerala, India, between 4th - 7th

INTERESTS AND EXTRACURRICULAR ACTIVITIES

- Trained in Yoga, Mrudangam, Football and was an integral part of the team winning 3rd in the intercollegiate MG University, Kerala football tournament.
- Traveling, Skywatching, Photography.
- Was part of the organising committee of a National level student conference, Young Astronomers Meet (YAM), organised by students, held in Physical Research Laboratory, Ahmedabad, during 24-28 September 2018.

PUBLICATIONS

Peer-Reviewed

1. **K Aravind**, Shashikiran Ganesh, Kumar Venkataramani, Devendra Sahu, Dorje Angchuk, Thirupathi Sivarani, Athira Unni, "*Activity of the first interstellar comet 2I/Borisov around perihelion: results from Indian observatories*", 2021, MNRAS, Volume 502, Issue 3, Pages 3491–3499.
2. **K. Aravind**, Prithish Halder, Shashikiran Ganesh, Devendra Sahu, Miquel Serra-Ricart, José J. Chambó, Dorje Angchuk, Thirupathi Sivarani, "*Optical observations and dust modelling of comet 156P/Russell-LINEAR*", 2022, Icarus, 383, 115042.
3. M. Bhatt, C. Wöhler, J. Rogall, **K. Aravind**, S. Ganesh and A. Bhardwaj, "*Unique regolith characteristics of the Reiner Gamma swirl as revealed by imaging polarimetry at large phase angles*", 2023, A&A 674, A82.
4. **Aravind K.**, Venkataramani K., Ganesh S., Jehin E., Moulane Y., "*Long-term spectroscopic monitoring of comet 46P/Wirtanen*", 2024, JApA, 45, 11.

5. **K. Aravind**, Kumar Venkataramani, Shashikiran Ganesh, Arun Surya, Thirupathi Sivarani, Devendra Sahu, Athira Unni, Anil Bhardwaj, “*Optical spectroscopy of comets using Hanle Echelle Spectrograph (HESP)*”, 2024, MNRAS, Volume 530, Issue 1, Pages 393–404.
6. J. Schmitt, C. Adami,...,**K. Aravind**, et al., “*Multi-purpose InSTRument for Astronomy at Low-resolution: MIS-TRAL@OHP*”, 2024, A&A, 687, id.A198, 17 pp
7. Shyama Narendranath, Shashikiran Ganesh, Dipen Sahu, Ritesh Kumar Mishra, Megha Bhatt, Sumanta Sarkhel, Guneshwar Thangjam, Kuljeet Kaur Marhas, **K. Aravind** et al., “*Solar system research prospects for the decade and beyond*”, 2025, JApA, Volume 46, article number 34.
8. S. Hmiddouch, E. Jehin, M. Lippi, M. Vander Donckt, **K. Aravind**, D. Hutsemékers¹, J. Manfroid, A. Jabiri, Y. Moulane and Z. Benkhaldoun, “*Variations in the volatile-driven activity of comet C/2017 K2 (PanSTARRS) revealed by long-term multiwavelength observations*”, 2025, A&A, 701, A61.
9. **K. Aravind**, E. Jehin, S. Hmiddouch, M. Vander Donckt, S. Ganesh, P. Rousselot, P. Hardy, D. Sahu, J. Manfroid and Z. Benkhaldoun, “*Ionic emission from and activity evolution in comet C/2020 F3 (NEOWISE): Insights from long-slit spectroscopy and photometry*”, 2025, A&A, 701, A161.
10. Goldy Ahuja, **K. Aravind**, Shashikiran Ganesh, Said Hmiddouch, Mathieu Vander Donckt, Emmanuel Jehin, Devendra Sahu, T Sivarani, “*Long-term monitoring of a dynamically new comet C/2020 V2 (ZTF)*”, 2025, MNRAS, Volume 543, Issue 2, Pages 1178–1195.
11. Rohan Rahatgaonkar, Juan Pablo Carvajal, Thomas H. Puzia, Baltasar Luco, Emmanuel Jehin, Damien Hutsemekers, Criel Opitom, Jean Manfroid, **K. Aravind** et al., “*VLT observations of interstellar comet 3I/ATLAS II. From quiescence to glow: Dramatic rise of NiI emission and incipient CN outgassing at large heliocentric distances*”, 2025, ApJL 995 L34.
12. Damien Hutsemékers, Jean Manfroid, Emmanuël Jehin, Cyrielle Opitom, Michele Bannister, Juan Pablo Carvajal, Rosemary Dorsey, **K. Aravind**, Baltasar Luco, Brian Murphy, Thomas H. Puzia, Rohan Rahatgaonkar, “*Extreme NiI/FeI abundance ratio in the coma of the interstellar comet 3I/ATLAS*”, 2025, Accepted for publication in A&A.
13. M. Vander Donckt, E. Jehin, **K. Aravind**, C. Adami, S. Hmiddouch¹, J. Manfroid, S. Ganesh, Z. Benkhaldoun, and A. Delsanti, “*Photometric and spectroscopic monitoring of the outbursting Halley-type comet 12P/Pons-Brooks*”, 2025, Accepted for publication in A&A.

Conference proceedings

14. Ganesh, Shashikiran, **Aravind Krishnakumar**, Kumar Venkataramani, Archita Rai, Kiran Singh Baliyan, and Umesh Chandra Joshi. “*Solar system studies with the Indo-Belgian telescopes.*”, 2019, Bulletin de la Société Royale des Sciences de Liège.
15. Ganesh, S., Rai, A., **Aravind K.**, Singh, A., Prajapati, P. V., Mishra, A., ... & Joshi, U. C., “*EMPOL: an EMCCD based optical imaging polarimeter.*”, 2020, In Ground-based and Airborne Instrumentation for Astronomy VIII (Vol. 11447, p. 114479E). International Society for Optics and Photonics.
16. Bhatt M., Wöhler C., **Aravind K.**, Ganesh S., Bharadwaj A., “*Regolith Characteristics of the Reiner Gamma Swirl as Revealed by Polarimetric Observations*”, 52nd Lunar and Planetary Science Conference, 2021, LPICo 2548, 2430.
17. Wöhler C., Bhatt M., Rogall J., **Aravind K.**, Ganesh S., Bhardwaj A., *Regolith properties of the lunar swirl Reiner Gamma inferred from phase ratio and polarimetric imaging*, 44th COSPAR Scientific Assembly, 2022, cosp, 44, 271.
18. **Aravind K.**, Venkataramani K., Ganesh S., Sahu D., Sivarani T., “*Revealing the ionic emissions in the comet C/2020 F3 (NEOWISE)*”, Asteroids, Comets, Meteors Conference, 2023, LPICo 2851, 2501.
19. M. Vander Donckt, **Aravind K.**, E. Jehin, S. Ganesh, S. Hmiddouch¹, Y. Moulane, Z. Benkhaldoun, A. Jabiri, D. Sahu, T. Sivarani, “*The Carbon-Chain depletion of recently observed Jupiter family comets from photometry and spectroscopy*”, Asteroids, Comets, Meteors Conference, 2023, LPICo 2851, 2461.
20. **Aravind K.**, Ganesh S., “*Optical spectroscopy of comets*”, BINA 2023, Bulletin de la Société Royale des Sciences de Liège (2023).
21. Wöhler C., Bhatt M., Arnaut M., Ganesh S., **Aravind K.**, A. Bharadwaj, “*Spectropolarimetric Properties of the Lunar Swirl Reiner Gamma and Western Oceanus Procellarum*”, 54th Lunar and Planetary Science Conference, 2023, LPICo 2806, 1609.
22. Adami, C.; Jehin, E.; **Aravind K.**; Ahmad, Y; Roy A.; Carvajal-Bohorquez C.; Garnichey M.; Hasan, M; Hassaine, D; Jimenez de la Vega, N. E.; Kaddour, A.; Kueviakoe, V.; Mendil, R.; Michel, F.; Anderson, S.; Delsanti, A.; LeCoroller, H.; Russeil, D; Lazzarini, S.; Litman, A. “*MISTRAL observations of the C/2022 E3 (ZTF) comet by the Aix-Marseille M2 students: First science results*”, SF2A-2023, Proceedings of the Annual meeting of the French Society of Astronomy and Astrophysics, 2023, pp.511-513.

23. Mathieu Vander Donckt, Emmanuel Jehin, **Aravind Krishnakumar**, Christophe Adami, Said Hmiddouch, Shashikiran Ganesh, Zouhair Benkhaldoun, Audrey Delsanti, and Abdelhadi Jabiri, “*Pre-perihelion TRAPPIST monitoring of the outbursting Halley-type comet 12P/Pons-Brooks*”, Europlanet Science Congress, Vol. 17, EPSC2024-951, 2024.
24. Said Hmiddouch, Emmanuel Jehin, Abdelhadi Jabiri, Youssef Moulane, **Aravind Krishnakumar**, Mathieu Vander Donckt, Goldy Ahuja, Zouhair Benkhaldoun, and Shashikiran Ganesh, “*Long-term TRAPPIST monitoring of a few Long Period and Dynamically New Comets*”, Europlanet Science Congress, Vol. 17, EPSC2024-1139, 2024.
25. **Aravind Krishnakumar**, Mathieu Vander Donckt, Youssef Moulane, Said Hmiddouch, Emmanuel Jehin, Goldy Ahuja, Shashikiran Ganesh, Devendra Sahu, Jabiri Abdelhadi, and Zouhair Benkhaldoun, “*Investigating the Molecular Activity and Composition of Comet C/2022 E3 via Photometry and Optical Spectroscopy*”, Europlanet Science Congress, Vol. 18, EPSC-DPS2025-1066, 2025.
26. Mathieu Vander Donckt, Emmanuel Jehin, **Aravind Krishnakumar**, Christophe Adami, Said Hmiddouch, Jean Manfroid, Shashikiran Ganesh, Zouhair Benkhaldoun, Audrey Delsanti, and Abdelhadi Jabiri, “*Photometric and spectroscopic monitoring of the outbursting Halley-type comet 12P/Pons-Brooks*”, Europlanet Science Congress, Vol. 18, EPSC-DPS2025-1646, 2025.
27. Goldy Ahuja, Shashikiran Ganesh, Agnes Fienga, **Aravind Krishnakumar**, and Zach Murra, “*N-Body Simulations of two Dynamically New Comets with different compositional characteristics*”, Europlanet Science Congress, Vol. 18, EPSC-DPS2025-1784, 2025.

Astronomers Telegram

28. **Krishnakumar, A.**, Angchuk, D., Venkataramani, K., et al. 2020, “*CN, C₂, C₃ production rates of Comet C/2020 F3 (NEOWISE) as observed from Himalayan Chandra Telescope, Hanle*”, India, The Astronomer’s Telegram, 13897, 1
29. Goldy Ahuja, **K. Aravind**, D. K. Sahu, Emmanuel Jehin, Mathieu Vander Donckt, Said Hmmdouch, Shashikiran Ganesh, Thirupathi Sivarani, 2024, “*Molecular gas production rates of Comet C/2023 A3 (Tsuchinshan - ATLAS)*”, India, The Astronomer’s Telegram, 16637, 1

REFERENCES

1) **Prof. Shashikiran Ganesh**
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2) **Prof. Emmanuel Jehin**
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 B-4000 Liège 1, Belgium
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